

## NETWORK SIMULATION LAB

### III B. TECH- II SEMESTER

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
| A5CS16      | PCC      | L            | T | P | C       | CIE           | SEE | Total |
|             |          | -            | - | 3 | 1.5     | 30            | 70  | 100   |

#### COURSE OBJECTIVES

1. Understand the importance and usage of NS3
2. To inculcate working knowledge among Node to Node communication in networks.
3. Understand and implement various networking topology.
4. To familiarize with the Net Anim

#### COURSE OUTCOMES

1. Install NS3 and its supporting packages.
2. Implement connecting various nodes
3. Implement Star and bus and Hybrid topologies using NS3
4. Use and develop applications using NetAnim.
5. Implement various routing protocols using NS3.

#### SYLLABUS

##### WEEK 1

Introduction about discrete events simulation and its tools.

##### WEEK 2

Installation of NS3 in Linux Operating System

##### WEEK 3

Program in NS3 to connect two nodes.

##### WEEK 4

Program in NS3 for connecting three nodes considering one node as a central node.

##### WEEK 5

Implement physical star topology and simulate using NS3.

##### WEEK 6

Implement physical bus topology and simulate using NS3.

##### WEEK 7

Program for connecting multiple routers and nodes and building a hybrid topology.

##### WEEK 8

Installation and configuration of NetAnim

##### WEEK 9

Program in NS3 to implement FTP using TCP bulk transfer

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| <b>WEEK 10</b>                                                                                                                                                                                                                                                                                   |  |
| Implement AODV Routing Protocol                                                                                                                                                                                                                                                                  |  |
| <b>WEEK 11</b>                                                                                                                                                                                                                                                                                   |  |
| Implement DSDV Routing Protocol.                                                                                                                                                                                                                                                                 |  |
| <b>WEEK 12</b>                                                                                                                                                                                                                                                                                   |  |
| Implement DSR Routing Protocol.                                                                                                                                                                                                                                                                  |  |
| <b>TEXT BOOKS</b>                                                                                                                                                                                                                                                                                |  |
| <ol style="list-style-type: none"> <li>1. Computer Networks, Andrew S Tanenbaum David J Wetherall , 5th Edition, Pearson Education.</li> <li>2. Computer Networking: A top down approach, Kurose, Ross (2010), Sixth Edition, Pearson Education, India.</li> </ol>                               |  |
| <b>REFERENCES</b>                                                                                                                                                                                                                                                                                |  |
| <ol style="list-style-type: none"> <li>1. Data communication and Networking, Behrouz A. Forouzan, 4th Edition, Mc Graw-Hill, India.</li> <li>2. Computer Networks: A Systems Approach, Larry L. Peterson, Bruce S. Davie, MK Publishers</li> </ol>                                               |  |
| <b>WEB LINKS</b>                                                                                                                                                                                                                                                                                 |  |
| <ol style="list-style-type: none"> <li>1. <a href="https://www.nsnam.org/documentation/videos/">https://www.nsnam.org/documentation/videos/</a></li> <li>2. <a href="https://www.nsnam.org/releases/ns-3-35/documentation/">https://www.nsnam.org/releases/ns-3-35/documentation/</a></li> </ol> |  |